

Predicting Customer Churn

A Data Engineering & Strategic Machine Learning Report

Kaggle Telecom Dataset Analysis | 100% Original Pipeline

Architecture

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Project Scope & Context

Auditing established datasets to build production-ready, defensible architectures from scratch.

| The Engineering Pipeline



Rigorous Cleanup

Fixed notebook noise and eliminated latent data leakage through strict temporal splitting and auditing features.



Original Codebase

Developed outside copy-paste environments. Every encoding decision and split strategy is original engineering work.



Deployment Ready

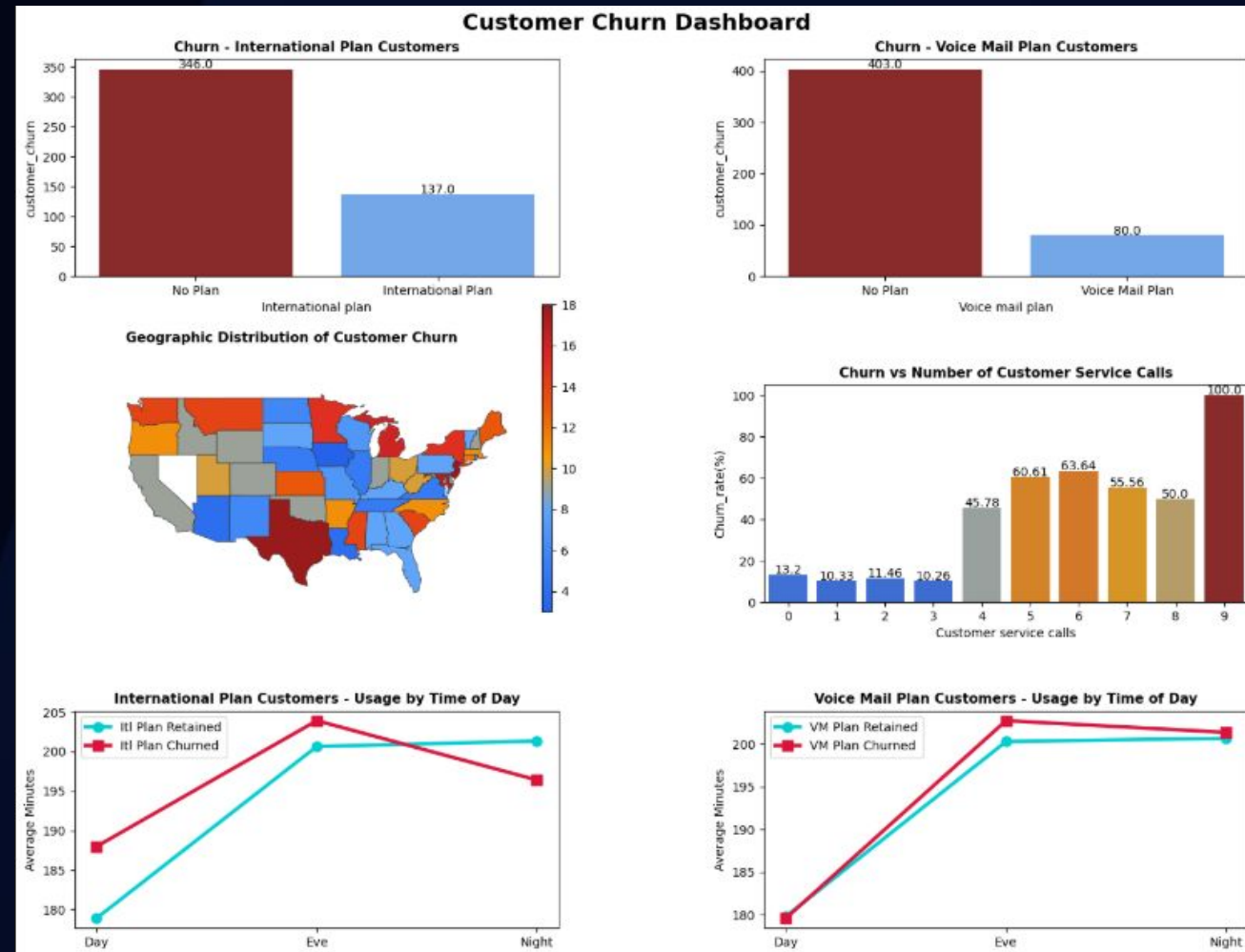
Designed with a "Production-First" mindset, ensuring stability against future drift in monthly data cycles.

Baseline Truth: Dashboard Analysis

6-Panel Matplotlib Grid

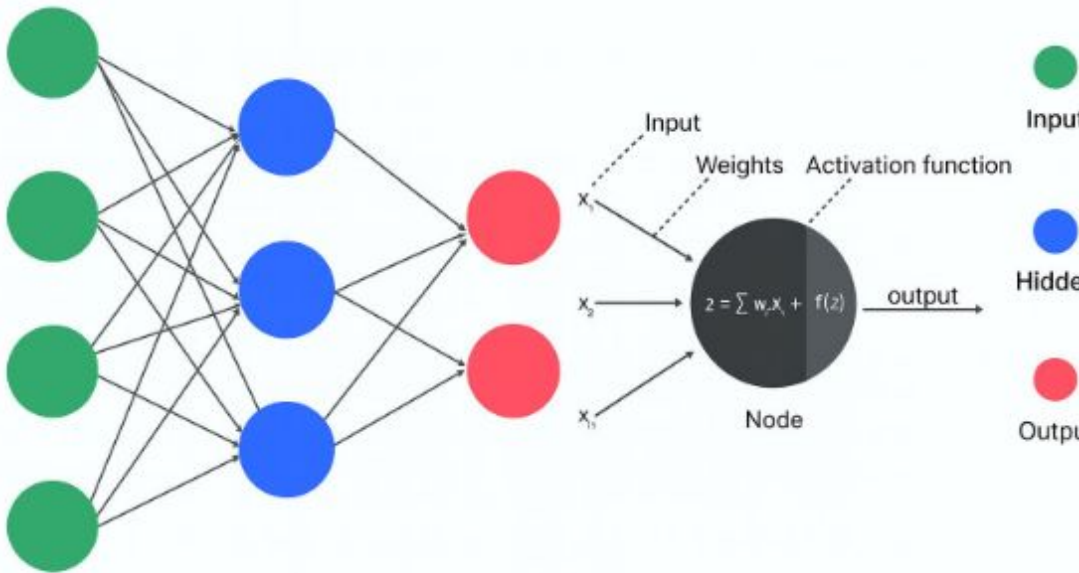
A coordinated, stateful visualization suite designed to reveal structural truths before modeling:

-  **Call Volume:** Sharp churn spikes after 3+ service calls.
-  **Geo-Choropleth:** State-level churn distribution mapping.
-  **Usage Curves:** Usage variations by Day/Eve/Night segments.
-  **Plan Metrics:** Comparative churn on Intl and VM plans.



Deep Learning: TensorFlow Experiment

Neural Network Architecture



Un-Guided Exploration

Tested Deep Learning boundaries in an open environment, focusing on raw network mechanics:



Cost Curves: Monitored convergence vs. overfitting real-time.



Imbalance Handling: Leveraged class weights to prioritize recall.



Trade-off: High recall (0.79) but surgical precision was sacrificed (0.41).

Model Performance Showdown

Model Architecture	Accuracy	Recall	Precision
Random Forest (RF)	0.93	0.57	0.92
XGBoost (Champion)	0.94	0.72	0.88
TensorFlow (TF)	0.81	0.79	0.41

XGBoost selected for its elite balance of precision and recall in production environments.

The Champion: XGBoost Pipeline

Linking data science insights back to the original baseline truth.

Feature Importance

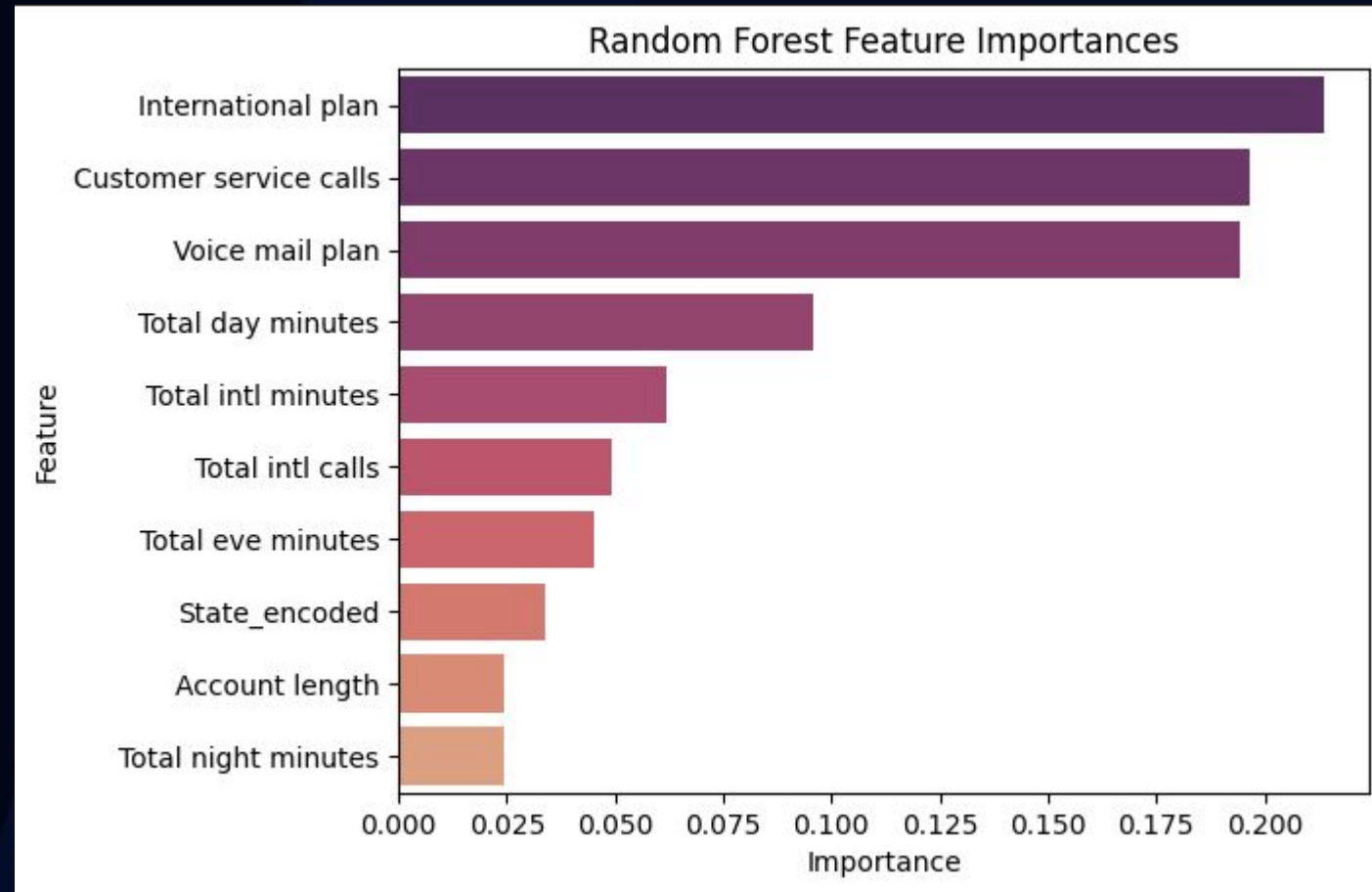
Linking to Baseline Truth

The model's weights directly validate our initial EDA findings:

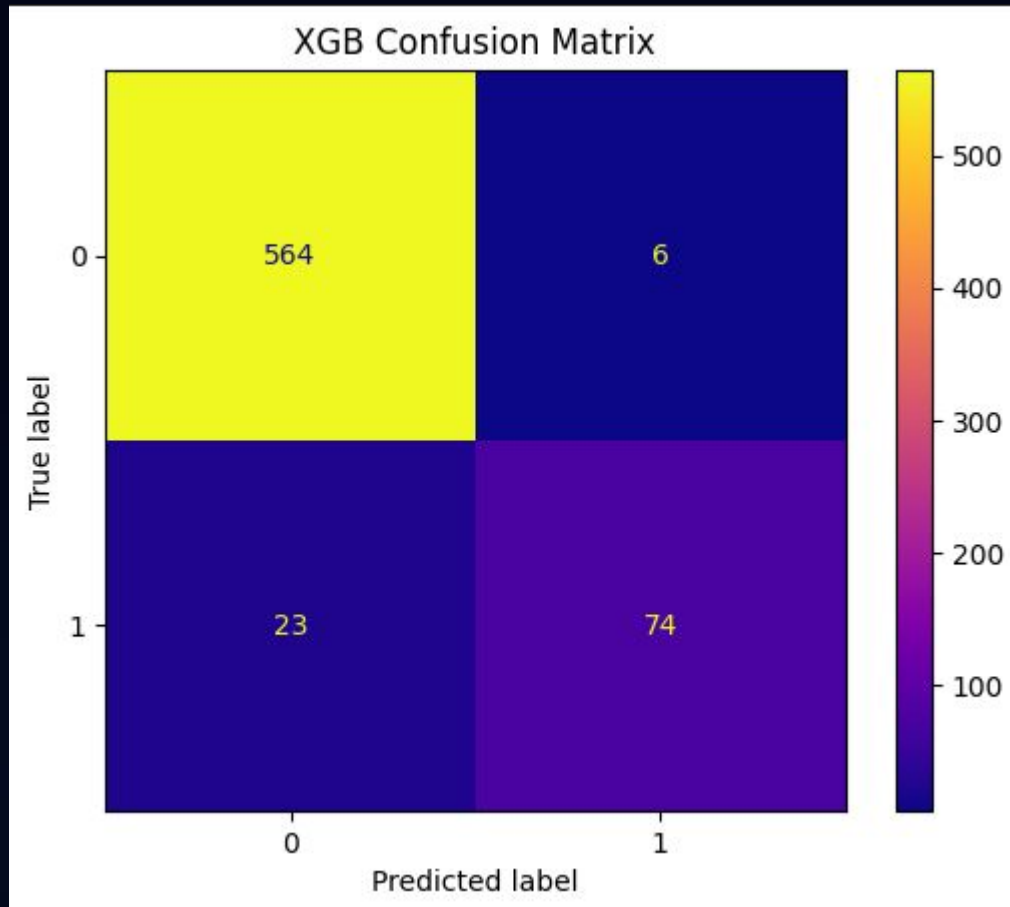
Customer Service Calls: Top importance, matching the "Danger Zone" identified in the dashboard.

International Plan: Heavy weight, aligning with high-churn usage patterns.

Total Day Minutes: Primary behavioral differentiator for attrition.



Champion Detail: Confusion Matrix



Predictive Accuracy

At the 0.5 decision threshold, the XGBoost model achieves surgical precision. By strictly controlling False Positives (Type I), we ensure business resources are only deployed on legitimate churn risks, maintaining long-term profitability.

| The Strategic Threshold Decision

0.5

PRODUCTION BASELINE

Why Reject the 0.3 Split?

Dropping to 0.3 offered a 4% recall bump but decimated precision to 0.79. We chose the stable 0.5 threshold to protect against **Future Data Drift**. A model optimized for current noise will collapse on next month's unseen distributions.

Business Impact & Capital Efficiency



The False Alarm Risk (Type I): Crying wolf leads to "Margin Erosion." Issuing coupons/incentives to happy customers who wouldn't have churned is a massive drain on operational efficiency.



Precision-Driven Retention: By maintaining 0.88 precision, we ensure retention efforts are laser-focused. XGBoost captures the majority of at-risk revenue without unnecessary spending.

Questions?

Predicting the future by engineering the present.

Thank you for your attention.

Portfolio: Churn Analysis Pipeline | 2026

Image Sources



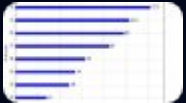
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